🍕 Pizza Sales Analysis : Customer Behaviour And Revenue Trends →

1. Project Overview :

The objective of this project was to analyze pizza sales performance using SQL and Excel. The project focuses on identifying revenue trends, top-performing products, customer Behaviour, and key business insights through data analysis and dashboard visualization.

2. Tools Used :

- MySQL
- Microsoft Excel
- Pivot Tables
- Pivot Charts
- Slicers

3. Dataset Information :

The dataset contains pizza sales transactions including:

- Order ID
- Order Date
- Order Time
- Pizza Name
- Pizza Category
- Pizza Size
- Quantity
- Total Price
- Pizza Ingredients

Additional helper columns created during analysis:

- Month
- Weekday
- Hour

4. Data Preparation :

Before performing the analysis, an Exploratory Data Analysis (EDA) process was conducted to understand the dataset and ensure data quality.

The following steps were performed:

- Reviewed the dataset structure and business context.
- Checked for duplicate records and verified data consistency.
- Validated and corrected data types for relevant columns.
- Created helper columns such as:
    - Month (for monthly trend analysis)
    - Weekday (for day-wise sales analysis)
    - Hour (for peak order hour analysis)
- Converted the raw dataset into an Excel Table format to improve data management and enable dynamic reporting.
- Performed preliminary data exploration to understand sales patterns and customer purchasing behavior.
- Created Pivot Tables to calculate key metrics and support dashboard development.
- Used Pivot Tables as the foundation for KPI calculations, business analysis, and visualization creation.

5. SQL Analysis :

SQL was used to answer key business questions and identify sales patterns within the dataset.

The following analyses were performed:

- Calculated Total Revenue generated by the business.
- Calculated Total Orders placed.
- Calculated Total Pizzas Sold.
- Identified the highest revenue-generating pizza category.
- Identified the highest revenue-generating pizza size.
- Determined the Top 10 revenue-generating pizzas.
- Determined the Bottom 10 revenue-generating pizzas.
- Analyzed revenue contribution by pizza category.
- Identified the pizza size with the highest quantity sold.
- Analyzed revenue performance by weekday.
- Analyzed order volume by hour.
- Determined the pizza category with the highest average order value.
- Compared category revenue against the overall average category revenue.
- Identified pizzas generating more revenue than the average pizza revenue within their own category.
- Ranked pizzas based on revenue within each category and identified the Top 3 revenue-generating pizzas per category.

SQL Concepts Used :

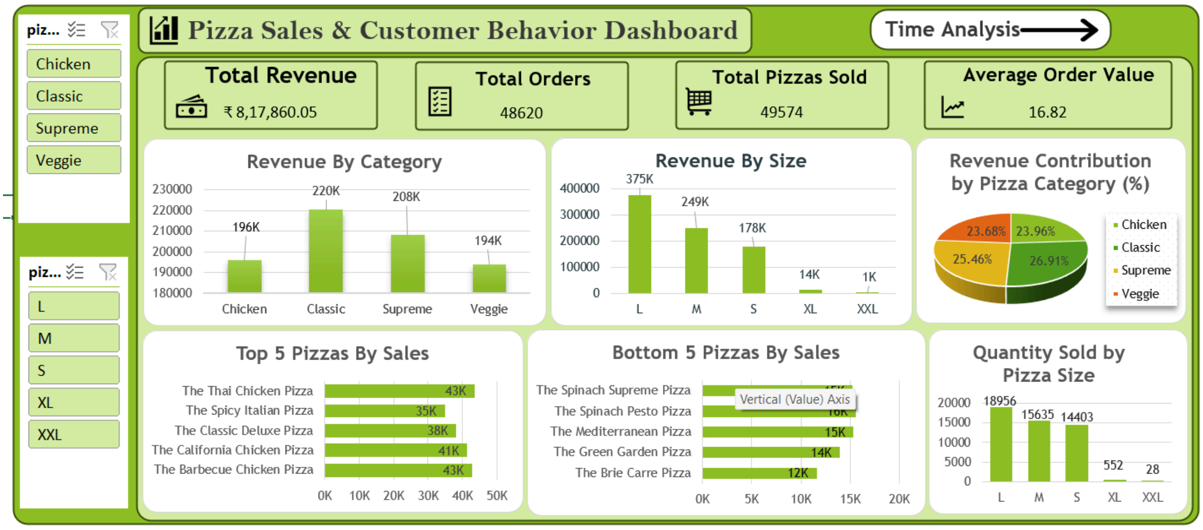
- Aggregate Functions (SUM, COUNT, AVG)
- GROUP BY
- ORDER BY
- LIMIT

- Common Table Expressions (CTEs)
- Subqueries
- Correlated Subqueries
- Window Functions
- RANK()
- PARTITION BY

6. Dashboard Development :

An interactive Excel dashboard was developed to visualize key business metrics and sales performance trends.

Executive Dashboard :



The Executive Dashboard provides a high-level overview of business performance through KPI cards and sales visualizations.

KPIs Included:

- Total Revenue
- Total Orders
- Total Pizzas Sold
- Average Order Value

Visualizations Included:

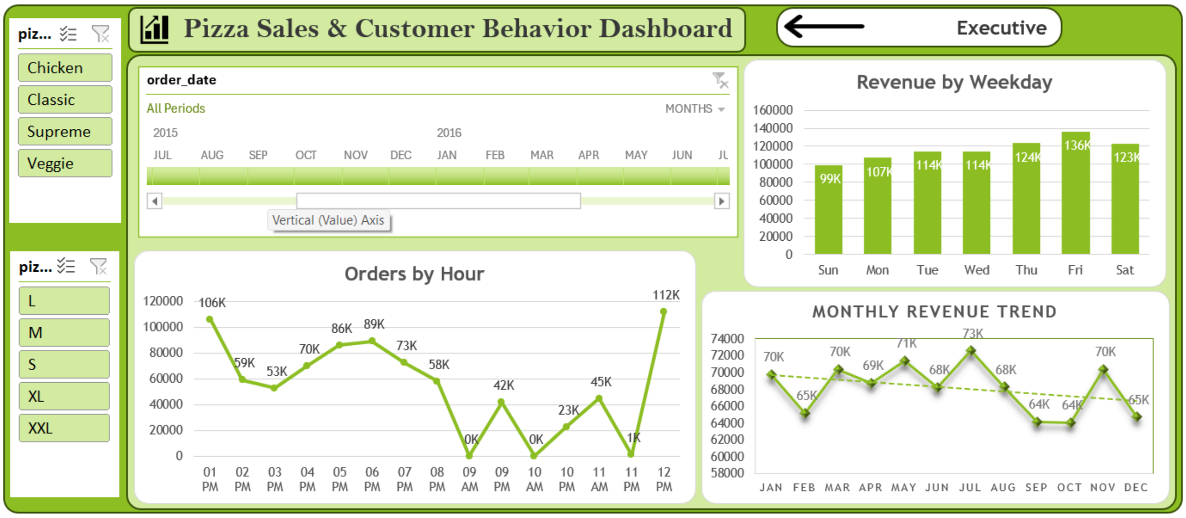
- Revenue by Pizza Category
- Revenue by Pizza Size
- Revenue Contribution by Category (%)

- Top 5 Pizzas by Revenue
- Bottom 5 Pizzas by Revenue
- Quantity Sold by Pizza Size

Interactive Features:

- Pizza Category Slicer
- Pizza Size Slicer
- Navigation Button to Time Analysis Dashboard

Time Analysis Dashboard :



A dedicated dashboard page was created to analyze sales trends and customer ordering behavior over time.

Visualizations Included:

- Monthly Revenue Trend
- Revenue by Weekday
- Orders by Hour

Interactive Features:

- Timeline Filter
- Navigation Button to Executive Dashboard

Dashboard Objective

The dashboard was designed to help users quickly monitor business performance, identify sales trends, analyze product performance, and explore customer ordering behavior through interactive filtering and visualization.

7. Key Insights :

- The business generated a total revenue of ₹817,860 from 48,620 orders and sold 49,574 pizzas during the analysis period.
- Classic pizzas generated the highest revenue (₹220K), contributing approximately 27% of total revenue among all pizza categories.
- Large-sized pizzas generated the highest revenue (₹375K) and also recorded the highest quantity sold (18,956 units), indicating strong customer preference for larger pizza sizes.
- The Thai Chicken Pizza was the highest revenue-generating pizza, contributing approximately ₹29K in sales.
- The Calabrese Pizza was among the lowest revenue-generating pizzas, generating approximately ₹6K in revenue.
- Friday generated the highest revenue (₹136K), while Sunday generated the lowest revenue (₹99K), indicating stronger customer demand toward the end of the week.
- Revenue peaked in July (₹73K), while September and October recorded the lowest monthly revenue levels at approximately ₹64K.
- Order activity was highest during the afternoon and evening hours, indicating peak customer demand during meal times.

8.Business Recommendations :

Based on the analysis and dashboard findings, the following recommendations can help improve business performance and support data-driven decision-making:

1. Promote High-Performing Categories

Classic pizzas generated the highest revenue among all pizza categories. The company can focus marketing campaigns, combo offers, and promotional activities around high-performing categories to maximize revenue and customer engagement.

2. Optimize Product Portfolio

Several pizzas generated significantly lower revenue compared to other products. The company should review the performance of low-selling pizzas and consider product improvements, targeted promotions, or menu optimization strategies to increase demand.

3. Focus on Large Pizza Sizes

Large-sized pizzas generated the highest revenue and recorded the highest quantity sold. The company can introduce family meal deals, bundle offers, and size-up promotions to further capitalize on customer preference for larger pizza sizes.

4. Leverage High-Revenue Days

Friday generated the highest revenue among all weekdays. The company can introduce special promotions, loyalty rewards, and targeted marketing campaigns on high-performing days to maximize revenue opportunities.

5. Improve Peak-Hour Operations

Order activity was concentrated during specific hours of the day. The company can allocate additional staff, inventory, and operational resources during peak periods to improve service efficiency and customer satisfaction.

6. Support Data-Driven Decision Making

The dashboard provides management with a centralized view of sales performance, product trends, and customer ordering behavior. Regular monitoring of these insights can support informed business decisions and continuous performance improvement.

9. Conclusion :

This project successfully analyzed pizza sales data using MySQL and Microsoft Excel to uncover key business insights and sales trends. Through SQL-based analysis, important metrics such as revenue, orders, product performance, category contribution, and customer ordering patterns were identified. An interactive Excel dashboard was developed to visualize KPIs, sales performance, and time-based trends, enabling users to explore the data through filters and interactive reports. The findings from this analysis can help businesses understand customer preferences, optimize product offerings, and support data-driven decision-making.